

Kmos-0140

Version: 1.0 | Status: Draft

SUMMARY

This specification defines the canonical schema for Assets within the Knowledge & Media Operating System (KMOS). Assets are the authoritative representations of digital evidence.

PURPOSE

This specification defines the canonical schema for Assets within the Knowledge & Media Operating System (KMOS). Assets are the authoritative representations of digital evidence.

Every file, document, recording, generated artifact, model output, publication package, and archival object SHALL be represented as a canonical Asset.

The Asset Schema establishes:

- Canonical Asset identity - Metadata - Lifecycle - Provenance - Lineage - Relationships - Governance - Validation - Conformance

This specification is implementation independent.

SCOPE

This specification applies to every persistent digital artifact managed by KMOS.

Examples include:

Video Audio Transcript Corrected Transcript Translation Subtitle Image Thumbnail Storyboard Presentation PDF Book Article Voice Recording Animation Knowledge Export Search Index Embedding Publication Package Configuration Snapshot Workflow Export Audit Report Evidence Package

Future asset types SHALL extend this specification rather than replacing it.

ASSET PHILOSOPHY

Assets represent evidence.

Knowledge Objects represent meaning.

Events represent change.

Capabilities produce and consume Assets.

Workflows coordinate Assets.

Applications present Assets.

Assets remain permanent records of institutional activity.

DESIGN PRINCIPLES

Every Asset SHALL be:

Canonically Identified Versioned Governed Traceable Reproducible Relationship-Aware Technology Independent Discoverable Validatable Preservable

Asset identity SHALL remain independent of storage technology.

CANONICAL ASSET STRUCTURE

Every Asset SHALL expose the following sections.

Identity: - Asset Identifier - Asset Type - Schema Version - Owner Domain - Lifecycle State

Metadata: - Display Name - Description - Media Type - Format - Language - Encoding - Content Length - Checksum - Hash Algorithm - Creation Timestamp - Modification Timestamp - Tags - Classification

Storage References: - Logical Storage Identifier - Physical Storage Location(s) - Replication Information - Archive Location - Content Address (if applicable)

Storage references SHALL NOT define Asset identity.

Governance: - Owner - Custodian - Approval Status - Review History - Security Classification - Retention Policy - Compliance Requirements - Certification Status

Relationships: - Parent Assets - Child Assets - Related Knowledge Objects - Related Events - Related Workflows - Related Publications - Related Capabilities

Related Assets SHALL reference canonical identifiers only.

CANONICAL ASSET IDENTITY

Every Asset SHALL possess a globally unique canonical identifier.

Identity SHALL remain stable across: - Storage migration - Cloud migration - Database migration - Application replacement - Filename changes - Directory changes

Identity SHALL survive for the lifetime of the Asset.

ASSET CATEGORIES

Canonical categories include:

Media Document Knowledge Representation Generated Artifact Workflow Artifact Configuration Publication Evidence Model Output Analytics Archive Operational Record

Additional categories MAY be introduced through future specifications.

ASSET METADATA

Metadata SHALL describe the business significance of the Asset rather than implementation details.

Recommended metadata includes: - Title - Description - Language - Media Characteristics - Duration - Resolution - Author - Creation Method - Content Classification - Subject - Associated Collection - Keywords - Institutional Owner

Metadata SHALL remain extensible.

PROVENANCE

Every Asset SHALL record its origin.

Required provenance information includes: - Original Source - Acquisition Method - Capture Time - Creation Method - Producing Capability - Producing Workflow - Producing Application - Human Contributors - AI Contributors - Review History - Approval History - Source Assets

Provenance SHALL explain why an Asset exists.

LINEAGE

Lineage records how Assets evolve.

Each Asset SHALL record: - Parent Assets - Input Assets - Derived Assets - Transformation Capabilities - Workflow Executions - Published Events - Approval Steps - Publication History

Lineage SHALL support complete reconstruction of Asset history.

ASSET LIFECYCLE

Assets progress through lifecycle states: Discovered → Registered → Validated → Processed → Reviewed → Approved → Published → Archived → Preserved → Retired

Lifecycle transitions SHALL generate canonical events.

Historical states SHALL remain queryable.

VERSIONING

Assets SHALL support immutable version history.

Each version records: - Version Identifier - Creation Event - Parent Version - Reason - Workflow - Capability - Approval - Supporting Evidence

Historical versions SHALL never be overwritten.

INTEGRITY

Every Asset SHALL support integrity verification.

Integrity metadata includes: - Checksum - Hash Algorithm - Verification Timestamp - Verification Method - Corruption Status - Repair History - Replication Status

Integrity SHALL be periodically revalidated according to governance policy.

RELATIONSHIPS

Assets participate in explicit relationships.

Examples: - Original Video contains Audio Track - Audio Track produces Transcript - Transcript supports Knowledge Object - Knowledge Object references Publication

Relationships SHALL themselves remain governable.

STORAGE INDEPENDENCE

Canonical Assets SHALL remain independent of storage technologies.

Implementations MAY use: - Object Storage - Distributed Storage - Local Filesystems - Network Storage - Cold Archives - Immutable Storage - Content Addressable Storage - Future storage technologies

Asset identity SHALL remain unchanged.

ASSET REGISTRY

Every Asset SHALL be registered within the Asset Registry.

Registry records include: - Identity - Metadata - Storage References - Lifecycle - Relationships - Lineage - Governance - Validation Status - Certification - Operational State

The Asset Registry is the authoritative catalog of digital evidence.

DISCOVERY

Assets SHALL support discovery through: - Identifier - Knowledge Object - Workflow - Capability - Publication - Language - Media Type - Topic - Contributor - Collection - Timeline

Search technologies remain implementation-specific.

GOVERNANCE

Every Asset SHALL record: - Institutional Owner - Custodian - Review History - Approval Status - Evidence Chain - Retention Rules - Security Classification - Compliance Records

Governance ensures institutional accountability.

VALIDATION

Assets SHALL pass validation before registration.

Validation includes: - Identifier Integrity - Checksum Verification - Metadata Completeness - Relationship Integrity - Storage Accessibility - Version Consistency - Governance Requirements

Validation failures SHALL publish canonical events.

REPRODUCIBILITY

Every published Asset SHALL be reproducible.

The platform SHALL preserve: - Input Assets - Workflow Version - Capability Versions - Configuration - Knowledge Version - Events - Approvals - Policies - Rendering Parameters

Reproducibility supports institutional trust and future verification.

CONFORMANCE

An implementation claiming compliance with KMOS-0140 SHALL: - Maintain globally stable Asset identifiers. - Support immutable version history. - Maintain provenance. - Maintain lineage. - Maintain governance metadata. - Support canonical lifecycle transitions. - Register Assets within the Asset Registry. - Publish canonical Asset events. - Support integrity verification. - Remain independent of storage technology.

Partial implementations SHALL document unsupported requirements.

RELATIONSHIP TO PREVIOUS SPECIFICATIONS

KMOS-0100 defined Canonical Objects.

KMOS-0110 defined Canonical Events.

KMOS-0120 defined Capability contracts.

KMOS-0130 defined Knowledge Objects.

KMOS-0140 defines the canonical representation of digital evidence supporting institutional knowledge.

Knowledge and Assets become complementary business entities.

Neither replaces the other.

ARCHITECTURAL CONSEQUENCES

Because every digital artifact is represented as a canonical Asset: - Storage technologies become interchangeable. - Applications no longer own files. - Capabilities automatically participate in lineage. - Knowledge remains linked to evidence. - Governance becomes comprehensive. - Historical reconstruction becomes possible. - Institutional preservation extends beyond storage into explainable evidence management.

REFERENCE EXAMPLE

A live lecture enters KMOS.

The platform creates: Lecture Video Asset → Audio Asset → Transcript Asset → Corrected Transcript Asset → Subtitle Asset → Storyboard Asset → Rendered Video Asset → Publication Package Asset → Archive Asset

Each Asset receives: - Canonical Identity - Metadata - Provenance - Lineage - Governance - Relationships - Lifecycle History

Although the physical files may reside in different storage systems over time, the canonical Asset identities remain unchanged and continue to provide the evidentiary foundation for the associated Knowledge Objects.

FUTURE ASSET SPECIFICATIONS

This specification defines the canonical Asset Schema.

Future specifications SHALL specialize this schema.

Examples include: - KMOS-1410 Media Asset Schema - KMOS-1420 Document Asset Schema - KMOS-1430 Publication Asset Schema - KMOS-1440 Archive Asset Schema - KMOS-1450 Configuration Asset Schema - KMOS-1460 Evidence Package Schema - KMOS-1470 AI Output Asset Schema - KMOS-1480 Preservation Package Schema

Each specialization SHALL extend this canonical schema while preserving interoperability.

NEXT SPECIFICATION

KMOS-0150 defines the Workflow Definition Language (WDL).

If: - KMOS-0100 defines canonical business objects, - KMOS-0110 defines canonical business events, - KMOS-0120 defines business capabilities, - KMOS-0130 defines institutional knowledge, - KMOS-0140 defines institutional evidence,

then KMOS-0150 defines the declarative language used to compose these components into repeatable, observable, governed business processes.

The Workflow Definition Language becomes the executable blueprint through which KMOS transforms knowledge, evidence, and capabilities into coordinated institutional operations.